

TABLE OF CONTENTS

E-103 GENERAOR CIRCUIT BREAKER

	Page No.
1. SCOPE OF WORK	E-103-2
2. CODES AND STANDARDS	E-103-2
2.1 National	E-103-2
2.2 International Electrotechnical Commission.....	E-103-2
2.3 American National Standards Institute.....	E-103-3
3. OPERATING AND DESIGN CONDITIONS	E-103-3
4. DETAILED REQUIREMENTS	E-103-3
4.1 General.....	E-103-3
4.2 Ratings.....	E-103-4
4.3 Insulating Medium.....	E-103-5
4.4 Operating Mechanism	E-103-6
4.5 Auxiliary Switches	E-103-7
4.6 Control Housing	E-103-7
4.7 Control Wiring.....	E-103-7
4.8 Accessories and Fittings	E-103-7
4.9 Porcelain	E-103-8
4.10 Nameplates	E-103-8
5. PERFORMANCE TESTING AND GUARANTEES	E-103-8
5.1 Type Test.....	E-103-8
5.2 Routine Test.....	E-103-9
6. QUALITY ASSURANCE/QUALITY CONTROL	E-103-9

E-103 GENERATOR CIRCUIT BREAKERS

1. SCOPE OF WORK

The Work shall include the design, supply, manufacturing, factory testing, delivery to site, installation, commissioning and testing of Generator Circuit Breakers in accordance with Part 4 Subsection 1.2 Summary of Work.

The scope of supply for each unit shall include, but not be limited to, the following:

- (1) Generator Circuit Breaker including isolator and earthing switches (2 each), voltage transformers, current transformers, surge arrester, disconnecting switch for connecting starting unit.
- (2) Operating mechanism
- (3) All instrumentation and controls

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of part 4, Section 2.5 Codes and Standards and the following specific standards:

2.1 National

- PUIL : Peraturan Umum Instalasi Listrik-Indonesia (General Regulation of Electrical Installation - Indonesia)
- SPLN : Standard Perusahaan Umum Listrik Negara National

2.2 International Electrotechnical Commission

- IEC 60060 : High-voltage test techniques.
- IEC 60137 : Bushings for alternating voltages above 1000 V.
- IEC 60265 : High-voltage switches.
- IEC 60376 : Specification and acceptance of new sulphur hexafluoride (SF₆).
- IEC 60417 : Recommended graphical symbols for use on equipment.
- IEC 60694 : Common specifications for high-voltage switchgear and control gear standards.

IEC 62155 : Hollow pressurized ceramic and glass insulators for use in electrical equipment with rated voltage greater than 1000 V.

IEC 62271 : High-voltage switchgear and control gear.

2.3 **American National Standards Institute**

ANSI/IEEE C37.013 : Standard for AC High-Voltage Circuit Breakers.
Rated on a Symmetrical Current

3. **OPERATING AND DESIGN CONDITIONS**

Circuit breakers shall be designed for indoor operation as a part of the Generator Isolated Phase Bus Duct System located in a tropical climate along the seacoast.

Each generator circuit breaker unit shall have a hydraulic spring operating mechanism for simultaneous 3-pole closing and tripping suitable for DC control. Electrically and mechanically of the operate mechanisms shall be provided completely with individual stored requirements to maintain service continuity.

The generator circuit breakers shall be of 100% self-extinguishing SF₆ type, designed for indoor service, metal-enclosed design, having electrically operated circuit breakers, three (3) pole operation, complying with IEC 62271 and ANSI/IEEE C37.013.

4. **DETAILED REQUIREMENTS**

4.1 **General**

The main current carrying parts shall be able to carry the rated current continuously without exceeding the temperatures specified in IEC Publication 62271 and 60265 as applicable.

Full contact and current carrying capacity shall be maintained during reasonable over travel and under travel of the mechanism.

The completely assembled circuit breaker shall be so designed that when the open circuit breaker is subject to impulse, 50 Hz, or switching surge

voltage (when a switching surge rating is called for in IEC 62271), the electric breakdown shall occur to ground and not across the open circuit breaker gap nor to adjacent phases.

Circuit breakers shall be capable of interrupting their rated interrupting current after trip coil has been energized within the specified time.

The contacts shall be self-wiping and self cleaning type.

4.2 Ratings

The Circuit Breakers, Disconnectors and Earthing Switches, complete with insulators and accessories, shall have the following ratings at the minimum. The rating shall be assigned on the basis of complete and adequate tests.

- (1) Nominal voltage.....: *) kV
- (2) Rated maximum voltage: *) kV
- (3) Frequency: 50 Hz
- (4) Number of poles: 3
- (5) Current ratings:
 - a. Continuous current: *) A
 - b. Short circuit breaking current: *) kA
 - c. Momentary current: *) kA
 - d. Short time (1 second) current: *) kA
- (6) Phase-to-ground insulation:
 - a. Impulse withstand voltage
(1.2 x 50 μ s) crest: *) kV
 - b. 50 Hz, dry (and wet*)
Withstand test voltage: *) kV
- (7) Open gap insulation:
 - a. Impulse withstand voltage
(1.2 x 50 μ s) crest: *) kV
 - b. One minute, 50 Hz, dry (and wet *)
withstand test voltage: *) kV

- (8) Rated interrupting time: 50 ms (2.5 cycle)
- (9) Rated permissible tripping delay: 1 second
- (10) CO-30 min - CO short circuit operation
CO-3 min - CO nominal load operation
- (11) First-pole-to-clear factor: 1.5

Calculations for determination of the rating shall be submitted for Owner's review and approval.

Note: *): Item to be determined by the Contractors and subject to approval by the Owner.

The breakers shall meet the Transient Recovery Voltage requirements specified in IEC 62271 (representation by four parameters) or equivalent for the rated short circuit conditions.

The generator circuit breaker shall be capable of synchronizing switching to the generator transformer.

Out of Phase Switching Requirement:

The reference restricting voltage test waveform shall be for a 25% breaker rating (ANSI/IEEE C37.013 or equivalent).

4.3 **Insulating Medium**

Circuit breakers shall be of the SF₆ insulating gas single pressure type.

The circuit breaker shall be provided with means to ensure that the moisture content, arc products, contaminants are maintained and filtered to prevent loss of dielectric or internal condensation on insulating surfaces.

Provisions to pull a vacuum of 8 millibars absolute or less, in less than four (4) hours and hold for required drying period prior to filling with insulating medium.

4.4 **Operating Mechanism**

The operating mechanism shall be of the pneumatic or hydraulic type.

The nominal control voltage for closing and tripping shall be 220 V DC. The ranges of the control voltages shall be 188 to 242 V DC for closing and shall be 154 to 242 V DC for tripping.

Circuit breaker operation:

For generator circuit breakers, operating mechanism shall be provided for three-pole operation. The operating mechanism shall have two (2) independent electric trip coils and dual pressure interlock switches. The two (2) trip coils including associated circuit and interlocks shall be wired independent and completely redundant.

Space heaters with necessary control devices shall be furnished and mounted in each mechanism housing.

The circuit breaker shall be operable (i.e. not in a lockout condition) after the completion of the operating duty cycle specified with the first opening operation occurring with the breaker pressure at the compressor starting pressure. Time delays shall be provided to prevent alarms due to pressure surges when operating breakers.

Alarm contacts shall be provided wherever failing of any heaters may result in faulty operation of the breaker or its components.

Contractor shall submit detailed Information for gas or air pressure levels in the breaker that the Contractor uses for closing, tripping, alarming or other functions.

A Manually emergency tripping device shall be provided, capable of tripping the circuit breaker during a complete loss of the AC and DC auxiliary power.

Circuit breaker mechanisms shall be provided with indicators or targets for each operating mechanism.

Operating mechanisms shall be provided with alarm contacts to close on low pressure and a low-pressure cut-out to prevent operation of the breaker closing valves with insufficient pressure.

Circuit Breaker shall be able to be operated from Local Control Panel as well as remotely from Central Control Room.

4.5 Auxiliary Switches

Auxiliary switches including spares shall be supplied with each operating mechanism.

4.6 Control Housing

Suitable cabinets of liberal size shall be provided for housing the operating mechanisms, relays and terminal blocks for the small wiring.

Hinged doors shall be provided on all cabinets that will allow easy access to all parts of the equipment mounted therein.

4.7 Control Wiring

Control wiring shall be in accordance with Technical Specification E-230.

External cables shall only enter the main breaker mechanism housing. The Contractor shall bring all connections to this mechanism housing.

4.8 Accessories and Fittings

Circuit breakers shall be equipped with a complete set of necessary accessories for operation and maintenance.

- (1) Position indicator.
- (2) Operation counter.
- (3) Means for effectively grounding tanks, mechanism housing, etc.
- (4) Maintenance closing device.
- (5) Pressure gauges.
- (6) Pressure switches.
- (7) Lifting devices.

- (8) Necessary interconnecting air and gas system piping.
- (9) Low voltage relays for monitoring the power supply to each compressor motor, heaters and closing and tripping circuits.
- (10) Tools for slow opening and closing of contacts for maintenance.
- (11) A complete charge of SF₆ gas for any gas filled portion of the circuit breaker.
- (12) Pressure switches to detect loss of SF₆ gas pressure or air pressure in any one of the three poles.

4.9 **Porcelain**

Contractor shall meet the requirements of IEC 62155 for all porcelain used in these circuit breakers. Porcelain insulation shall be chocolate brown color and shall be permanently marked.

4.10 **Nameplates**

Nameplates shall be as per IEC 62271. The compressor and tank for the operating mechanism shall have nameplates giving normal and maximum pressure rating.

5. **PERFORMANCE TESTING AND GUARANTEES**

5.1 **Type Test**

Normal type tests as per IEC 62271 and 60265 or equivalent as applicable shall be performed on a sample of each type of circuit breakers and the test results shall be submitted.

Special type tests in addition to those outlined in IEC 62271, 60265 and as per ANSI/IEEE C37.013 or equivalent shall be performed as applicable as described herein.

- (1) No load timing tests as per IEC 62271 shall be supplemented by the following tests and shall meet the same timing requirements.
 - a. CO duty cycle at lockout pressure and minimum coil voltage.
 - b. O-t-CO auto reclose cycle at lockout pressure and minimum coil voltage
- (2) Tests shall be performed on a sample of each type of circuit breaker

for verifying insulation coordination of the open gap and to ground for critical flashover impulse voltage using the IEC 60060 “up and down” method.

- (3) One (1) of each container type subject to impulse pressure shall be tested to three (3) times maximum impulse pressure during short circuit conditions without a permanent distortion. Measurements of temporary distortion during test shall also be made.

Incase the above type test already performed on identical equipment, than the type test report shall be submitted. If the type test report is accepted by The Owner than no repeated type test need to be done, if rejected than type test must be done.

5.2 Routine Test

Routine test as per IEC 62271 shall be carried out as outlined below.

- (1) Power frequency, voltage dry tests of the main circuit of each voltage level shall be performed on 10% of each design units with a minimum of two (2) units tested.
- (2) Voltage tests on control and auxiliary circuits shall be performed on all units.
- (3) Measurement of the resistance of the main circuit samples are same as clause (1) above.
- (4) Mechanical operating tests shall be performed on all units.
- (5) Mechanical/ hydraulic pressure tests on all containers under pressure to twice the maximum operating pressure for static pressures and three (3) times the maximum operating pressure for containers subject to impulse or rapid pressure changes.

6. QUALITY ASSURANCE/QUALITY CONTROL

The quality assurance/quality control for generator circuit breaker shall conform to the requirements included in Part 4, Section 2, subsection 2.7. The procedures specified shall include, but not be limited to, the followings:

- (1) Power frequency voltage withstands tests.

- (2) Voltage tests on wiring.
- (3) Resistance measurement tests on main circuit.
- (4) Mechanical operating and pressure tests.

The documentation shall include, but not be limited to, the following:

- (1) Shop Test Reports
- (2) Operating tests to check speed and mechanical performance.
- (3) Storage tank leakage tests.

(Last page of E-103)